GEBZE TECHNİCAL UNİVERSİTY

Faculty of Sciences · Department of Data Science and Analytics

DSA107 — DATA SCİENCE AND ANALYTİCS LABORATORY

LABORATORY REPORT No. 2

Summarising One Variable, Then Two

A descriptive and cross-tabular analysis of the Titanic passenger dataset

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1. Objective

The objective of this study is to summarise categorical and quantitative variables from the Titanic passenger dataset, moving from single-variable descriptions to two-variable relationships. The primary goal is to distinguish stored data types from true variable types, summarise a categorical variable using counts and proportions, and cross-tabulate two categorical variables into conditional rates to evaluate survival differences fairly across groups of unequal size. Additionally, the study aims to describe quantitative variables using histograms, measures of centre (mean and median), and spread (standard deviation) to assess skewness and extreme values.

2. Data Description

The dataset analysed is a version of the Titanic passenger list, retrieved directly from the Seaborn data repository via URL using pd.read_csv. It contains 891 rows and 15 columns, where each individual row represents a single passenger on board. The columns include categorical identifiers (survived, pclass, sex, embarked, class, who, adult_male, deck, embark_town, alive, alone) and quantitative measurements (age, sibsp, parch, fare). Although pclass and survived are stored as integer data types (int64), they represent categorical labels rather than numerical amounts.

Missing values are concentrated in four columns: deck has 688 missing entries (77.2%), age has 177 (19.9%), while embarked and embark_town miss 2 entries each (0.2%). The remaining 11 columns are complete. Due to the severe lack of data in deck (over three-quarters missing), this column was excluded from further analysis to avoid biased interpretations. For age and fare summaries, calculations automatically handle or ignore missing entries without requiring row deletion. Table 1 provides an overview of the dataset structure.

Property	*Value*
Source	Seaborn repository (mwaskom/seaborn-data/master/titanic.csv), loaded via URL
Size	891 rows x 15 columns
Row Meaning	One passenger on board the Titanic
Key Categorical Columns	pclass, survived, sex, deck, embark town
Key Quantitative Columns	age, fare
Missing values	Deck: 688 (77.2%), age: 177 (19.9%), embarked / embark _town: 2 (0.2%)
Rows used	All 891 rows kept; missing values were handled automatically by functions without dropping rows.

Table 1. Overview of the dataset.

3. Methodology

All analytical steps were performed in a single Google Colab notebook using Python 3. The pandas library was utilized for data manipulation and cross-tabulation, while matplotlib and seaborn were used for visualization. After reading the dataset directly from its URL via pd.read_csv, the initial inspection was conducted using .head() to verify correct loading, followed by .shape, .info(), and .isna().sum() to determine the dataset's dimensions, column types, and the distribution of missing values across features.

To summarise the categorical variable pclass, .value_counts() was applied to extract both raw counts and relative proportions. For the relationship between passenger class and survival, pd.crosstab was used. First, marginal row totals were computed to find the total passengers per class. Then, the table was converted into a conditional distribution (rates) by normalizing across the index (rows). This step is crucial because dividing each cell by its own row total allows for a fair comparison of survival probabilities between classes of different sizes.

For the quantitative variables, .plot.hist(bins=10) was used to visualise the distribution of ages by grouping values into distinct bins. Summary statistics (.mean(), .median(), .std(), and .max()) were calculated for age and fare to identify central tendencies and spread. Finally, the notebook was executed top-to-bottom using "Restart and run all" to ensure reproducibility, and the completed code was pushed to the GitHub repository.

4. Results

The dataset contains 891 passengers, with missing values distributed unevenly across the columns. The deck column contains the highest number of missing values (688), followed by age (177), while embarked and embark_town each lack only 2 values.

When examining the categorical distribution of passenger classes, the data shows that more than half of the passengers travelled in third class. Specifically, there are 491 third-class passengers, making up 55.1% of the dataset. First class contained 216 passengers (24.2%), and second class had 184 passengers (20.7%). The raw counts for each class are visualised in Figure 1, which clearly shows that third class is the largest group.

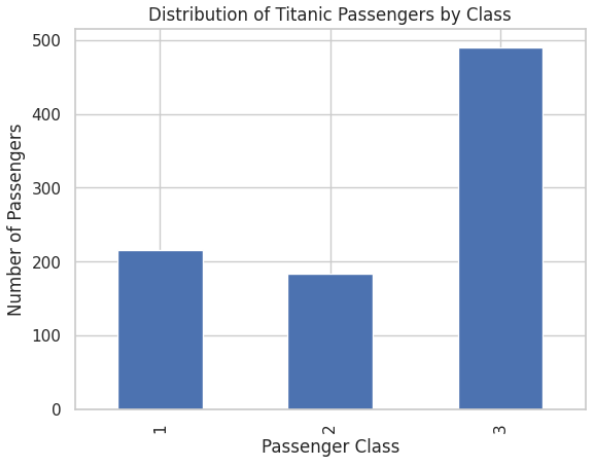


Figure 1. Distribution of Titanic passengers by socio-economic class

To determine how class related to survival, a cross-tabulation was performed. The raw joint counts indicate that third class had the highest number of deaths (372 passengers). However, converting these counts into a conditional distribution (rates normalized by row totals) provides a proportional view of survival. As seen in Figure 2, only 24.2% of third-class passengers survived, whereas 63.0% of first-class passengers and 47.3% of second-class passengers survived.

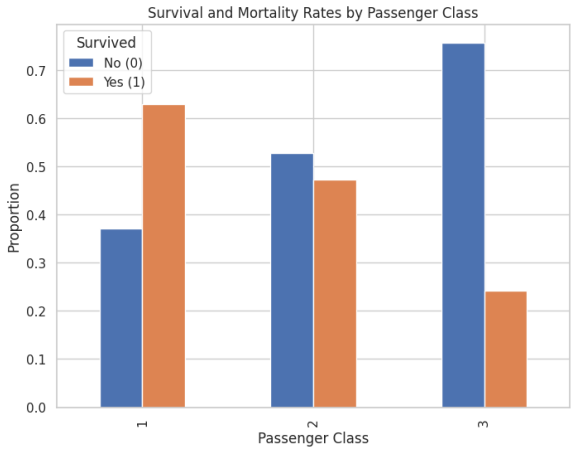


Figure 2. Survival and mortality rates normalized by passenger class.

For the quantitative variables, summary statistics are presented in Table 2. The mean age of the passengers is 29.70 years, with a median of 28.00 and a standard deviation of 14.53. The age distribution was further grouped into 10 bins, as shown in Figure 3. The histogram reveals that the largest group of passengers is concentrated between 20 and 30 years of age. Regarding ticket prices, the mean fare is 32.20, but the median fare is significantly lower at 14.45, while the maximum fare reaches up to 512.33.

Column	Count	Mean	Median	Std	Min	Max
age	714	29.70	28.00	14.53	0.42	80.00
fare	891	32.20	14.45	49.69	0.00	512.33

Table 2. Summary statistics of the key quantitative columns, from .describe().

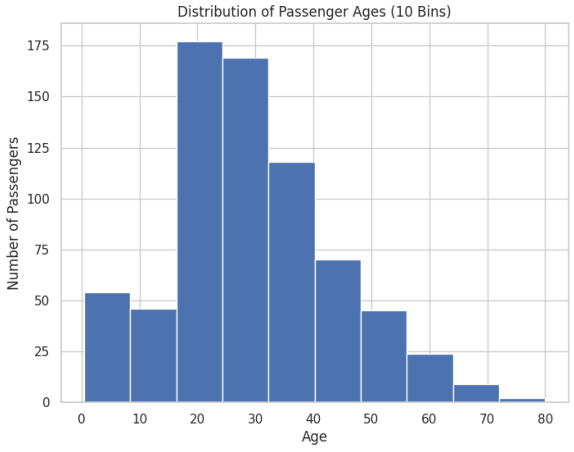


Figure 3. Distribution of passenger ages grouped into 10 bins.

5. Discussion

Conditional rates reveal a higher survival probability for first-class (63.0%) versus third-class (24.2%) passengers. This demonstrates that using rates, rather than raw counts, is essential to compare unequal groups fairly. However, this statistical link does not prove causality; a multi-variable analysis including gender is required next.

Second, handling missing data requires column-specific decisions. Blindly removing rows for the deck column (77.2% missing) would wrongly discard most valid passenger records.

Finally, extreme outliers pull the fare mean (32.20) far above the median (14.45). This right-skewness shows why the median is a more honest summary of typical prices.

6. Reflection

Navigating Google Colab and GitHub was noticeably easier than last week, though I am still in the adaptation process and expect to improve my workflow further. The most challenging part of this lab was understanding which denominator to use in cross-tabulations. I overcame this by calculating the row totals first to see how groups differed in size. The most valuable lesson I learned is that raw counts can be highly misleading; using conditional rates is essential for fair comparisons. Additionally, I now understand why the median is a much safer metric than the mean when data is skewed by outliers.

7. References

Waskom, M. (2021). seaborn: statistical data visualization. Journal of Open Source Software, 6(60), 3021.
Retrieved from https://raw.githubusercontent.com/mwaskom/seaborn-data/master/titanic.csv
Notebook repository: https://github.com/orhanunser/dsa107-labs.git

AI Assistance Disclosure

Tools used: *Google Gemini*.
Purpose: Structuring the report format, improving academic English, and ensuring compliance with the lab guidelines. I will use it less and less in incoming weeks. All code was written, run, and verified by me, and every number in this report was read directly from my own notebook output.